

Vitamin and Mineral Health Performance, Health Claims, and Research Findings

Vitamin	Proposed Ergogenic Claim	Summarized Research Findings
Provitamin A (Carotenoids)	Antioxidant: ↓ exercise-induced lipid peroxidation and exercise induced-muscle damage	No protection against exercise-induced lipid peroxidation and inflammation and may hinder the recovery of muscle damage (Teixeira et al., 2009).
B₁ (Thiamin)	Energy system: ↑ anaerobic threshold and CO ₂ transport.	No effect when athletes consume normal intake. A deficiency could ↓ enzyme activation; ↓ ATP biosynthesis that leads to earlier onset fatigue (Choi et al., 2013; Kerksick et al., 2018).
B₂ (Riboflavin)	Energy system: ↑ energy availability during oxidative metabolism (aerobic exercise)	No effect on aerobic performance (Gonçalves & Portari, 2021; Kerksick et al., 2018).
B₃ (Niacin)	Substrate utilization, reduce cholesterol, energy systems: ↑ in fatty acids during exercise, ↓ cholesterol, ↑ thermoregulation, and ↑ substrate availability during oxidative metabolism	Supplementation (100–500 mg·d ⁻¹) can help ↓ blood lipid levels in persons with hypercholesteremia; ↓ exercise capacity by blunting the mobilization of fatty acids (Kerksick et al., 2018).
B₅ (Pantothenic acid)	Energy systems: Used to make Acyl-CoA and may benefit aerobic or oxygen energy systems.	No improvement on aerobic performance; may influence lactate accumulation (Kerksick et al., 2018).
B₆ (Pyridoxine)	Muscle adaptations and energy systems: ↑ muscle mass, strength, and aerobic power in the glycolytic and aerobic energy systems.	No effect, did improve aerobic capacity, or slow lactate accumulation (Kerksick et al., 2018).
B₉ (Folic acid)	Oxygen capacity: ↑ in RBCs and ↑ oxygen delivery to the muscles during exercise	No deficiency, no improve exercise performance (Kerksick et al., 2018).
B₁₂ (Cobalamin)	Muscle adaptations and oxygen capacity: ↑ muscle mass, and ↑ RBCs to ↑ the oxygen-carrying capacity of the blood	No deficiency, no effect on exercise performance (Kerksick et al., 2018).
Vitamin C (Ascorbic acid)	Exercise performance, antioxidant, and immunity: ↑ metabolism during exercise and enhance immunity	No deficiency, no effect on exercise performance; some data to support that 500 mg·d ⁻¹ post high-intensity exercise bout may ↓ the incidence of upper respiratory tract infections; supplemental vitamin C has shown positive, negative, and negligible impact on muscle performance markers; high doses should be minimized in they may negatively affect resistance training adaptations (Deane et al., 2017; Kerksick et al., 2018; Paulsen et al., 2014).

Vitamin and Mineral Health Performance, Health Claims, and Research Findings

Vitamin	Proposed Ergogenic Claim	Summarized Research Findings
Choline*	Neuromuscular fatigue: ↑ choline availability would lead to delay in fatigue	Long exhaustive exercise did not deplete plasma choline nor did supplementation of choline delay fatigue (Warber et al., 2000).
Vitamin A (Retinol, retinal, and retinoic acid)	Vision performance: ↑ in vision performance	No studies have shown any benefit to exercise performance with vitamin A supplementation (Kerksick et al., 2018).
Vitamin D (D ₂ - ergocalciferol and D ₃ - cholecalciferol) Provitamin (7-dehydrocholesterol)	Muscle growth and bone mineralization: ↑ in muscle growth and bone mineral content, ↑ adaptive response to exercise, ↓ stress fractures	Muscle and strength increase in elderly or deficient populations; supplementation may reduce incidence of stress fractures; no effect on exercise performance (Deane et al., 2017; Kerksick et al., 2018; Maughan et al., 2018).
Vitamin E (Tocopherols and tocotrienols)	Antioxidant: ↑ oxygen delivery to the muscles during exercise	Vitamin E supplementation can decrease exercise-induced oxidative stress; may hinder exercise-induced mitochondrial biogenesis and antioxidant enzymes in skeletal muscle; free radicals may be an important signaling agent for resistance training muscular adaptations (Deane et al., 2017; Kerksick et al., 2018; Paulsen et al., 2014).
Vitamin K (Phylloquinones and menaquinones)	Bone metabolism: ↑ in bone density	An increase in bone formation markers in females (Kerksick et al., 2018).

The up arrow (↑) denotes an increase, and the down arrow (↓) denotes a decrease. Adapted from Kerksick et al. (2018) and Maughan et al. (2018) for claims that may be more related to physique athletes and competition preparation.

Mineral	Proposed Ergogenic Claim	Summarized Research Findings
Calcium (Ca)	Bone metabolism and body composition: ↑ fat metabolism. Diet should contain sufficient amounts in female athletes and postmenopausal women.	No ergogenic effect on exercise performance; supplementation may be beneficial for females at risk for osteoporosis; dairy products and calcium intake may be factors in regulating fat metabolism (Kerksick et al., 2018; Zemel, 2003).
Phosphorus (P)	Energy system: ↑ availability for the ATP-PCr, glycolytic, and aerobic energy systems	Shown to improve aerobic capacity in endurance athletes at 3-5 g·d ⁻¹ ; unknown if any benefit with resistance training (Jeukendrup & Gleeson, 2018).
Potassium (K)	Electrolyte: ↑ in potassium may ↓ exercise-associated muscle cramping (EAMC)	No ergogenic effect; EAMC occur in a variety of conditions, environmental conditions, and populations; muscle cramps may be multifaceted (dehydration, electrolyte imbalance, or neuromuscular factors) (Kerksick et al., 2018; Miller et al., 2010).

Vitamin and Mineral Health Performance, Health Claims, and Research Findings

Mineral	Proposed Ergogenic Claim	Summarized Research Findings
Sodium (Na)	Plasma volume and diuresis: ↑ in Na ingestion leads to ↑ blood volume and increases diuresis and fluid loss prior to competition	Pre-exercise Na ingestion may be an ergogenic aid for subsequent prolonged aerobic exercise in a warm or thermoneutral environment; no association between the increase in Na intake and urine volume. Extra fluid ingested remains in the body to dilute the Na concentration that cannot be excreted quickly enough by kidneys; it takes up 2 to 3 days to create a new steady state of Na intake to excretion (Bankir et al., 2017; Mora-Rodriguez & Hamouti, 2012).
Magnesium (Mg)	Energy system: ↑ improve ATP synthesis and availability	No deficiency, no effect on exercise performance; some positive changes occurred with supplementation; previous studies did not clearly indicate that magnesium supplementation should be considered an ergogenic aid (Kerksick et al., 2018; Volpe, 2015).
Iron (Fe)	Aerobic capacity: ↑ aerobic performance in activities that recruit the aerobic energy system	No deficiency, no ergogenic effect, does not appear to improve aerobic performance unless individual is iron-depleted and/or has iron deficiency-anemia (Kerksick et al., 2018; Rodenberg & Gustafson, 2007).
Zinc (Zn)	Immune system: ↓ risk of infection	No effect on exercise performance; evidence to suggest when cold symptoms begin, taking zinc lozenges (>75 mg zinc·day ⁻¹) can reduce the number of days illness/symptoms (Gleeson & Williams, 2013; Kerksick et al., 2018; Volpe, 2008).
Selenium (Se)	Antioxidant: ↑ aerobic exercise performance, may assist in reducing free radical production of lipids during aerobic exercise	No effect on exercise performance; may reduce lipid peroxidation during aerobic exercise (Kerksick et al., 2018; Volpe, 2008).
Chromium (Cr)	Insulin sensitivity and body composition: ↑ lean body mass and ↓ body fat levels	Animal models show an augmented insulin signaling and lean body mass gain. These outcomes not seen in human models (Hua et al., 2012; Kerksick et al., 2018)

The up arrow (↑) denotes an increase, and the down arrow (↓) denotes a decrease. Adapted from Kerksick et al. (2018) and Maughan et al. (2018) for claims that may be more related to physique athletes and competition preparation.